

CHUQUR O'RGANISH USULLARI · Amaliy mashg'ulot (Practice)

7-amaliy mashg'ulot. Umumlashtirish, regularizatsiya va model tanlash

2026–2027 o'quv yili · 30 ta topshiriq

Mavzu. Umumlashtirish farqi, bias–variance, L1/L2 regularizatsiya va weight decay (AdamW), dropout, data augmentation, early stopping, batch normalization, He/Xavier initialization, learning-rate schedules (warmup, cosine annealing), transfer learning, ansambllar, giperparametr qidiruvi va ma'lumot sizishi (data leakage).

Maqsadlar. Talaba: (1) regularizatsiya formulalarini qo'lda hisoblaydi va keltirib chiqaradi; (2) PyTorch va scikit-learn API larining aynan qanday hisoblashini tekshiradi; (3) kichik, lekin real eksperimentda (shovqinli yorliqli `load_digits`) usullarni bir xil sharoitda taqqoslaydi va to'g'ri xulosa chiqaradi.

Baholash (100 ball). A bo'lim (1–12): $12 \times 2 = 24$; B bo'lim (13–16): $4 \times 1,5 = 6$; C bo'lim (17–30): $14 \times 5 = 70$. Kodli topshiriqda: to'g'ri va takrorlanuvchi natija — 3 ball, kod sifati (imzo, urug', `model.eval()`, sizishsiz) — 1 ball, natijaning izohi — 1 ball. Topshirish: bitta `solutions.py` (har bir kodli topshiriq — alohida funksiya va `main`) va hisobot (PDF).

Muhitni sozlash (Python 3.11 yoki 3.12, faqat CPU)

```
python -m venv .venv; source .venv/bin/activate (Windows: .venv\Scripts\activate); pip
install -r requirements.txt. Versiyalar qat'iy belgilangan: numpy==2.4.4, scipy==1.17.1,
pandas==3.0.2, scikit-learn==1.8.0, scikit-image==0.26.0, matplotlib==3.10.9, pillow==12.2.0,
opencv-python-headless==4.13.0.92, torch==2.14.0, torchvision==0.29.0. Ishga tushirish: python -W
error::DeprecationWarning solutions.py — ogohlantirishsiz ishlashi shart. Barcha ma'lumotlar offlayn:
sklearn.datasets.load_digits (1797 ta  $8 \times 8$  tasvir), load_breast_cancer ( $569 \times 30$ ) kutubxona ichida;
qolganlari sintetik. Urug' 42, torch.set_num_threads(1). Umumiy runtime — 3 daqiqadan kam.
```

A. Qo'lda hisoblash va keltirib chiqarish (1–12)

- Umumlashtirish farqi va ishonch oralig'i.** O'quv aniqligi 0,975, test aniqligi $\hat{a} = 0,912$, $n_{\text{test}} = 2000$. Umumlashtirish farqini, $\text{SE}(\hat{a})$ ni va 95% ishonch oralig'ini toping. Ikkinchi model 0,918 berdi: farq statistik ahamiyatlimi?
- Bias–variance.** $y = f(x) + \xi$, $\text{Var } \xi = \sigma^2 = 0,25$, $f(x_0) = 2$. Besh xil o'quv to'plamida o'qitilgan model x_0 da 1,6; 2,2; 1,9; 1,5; 1,8 ni bashorat qildi. bias^2 , variance va kutilgan kvadratik xatoni hisoblang.
- Ridge (L2).** $\min_w \|y - wx\|^2 + \lambda w^2$ uchun $w_\lambda = \frac{x^T y}{x^T x + \lambda}$ ni keltirib chiqaring. $x = (1, 2, 2, 3)$, $y = (1, 3, 2, 4)$ da w_0 , w_2 , w_{18} ni toping.
- L1 va L2 proksimal qadam.** $z = (1, 2; -0,4; 0,25; -0,9)$, $\lambda = 0,5$. $\min_w \frac{1}{2}(w - z)^2 + \lambda|w|$ va $\min_w \frac{1}{2}(w - z)^2 + \frac{\lambda}{2}w^2$ yechimlarini koordinatalar bo'yicha yozing. Qaysi birida nollar paydo bo'ladi?
- Label smoothing.** $K = 5$, logitlar $z = (2, 1, 0, 0, -1)$, to'g'ri sinf 0. Softmax ehtimolliklarini, oddiy cross-entropy ni va $\varepsilon = 0,1$ li label smoothing nishoni $y^{\text{LS}} = (1 - \varepsilon)\mathbf{e}_0 + \varepsilon/K$ bilan cross-entropy ni hisoblang (4 xona).
- Inverted dropout.** $p = 0,25$, $h = (2, 0, -1, 4)$, niqob $b = (1, 0, 1, 1)$. $\tilde{h}$ ni toping. $\mathbb{E}[\tilde{h}_j | h_j] = h_j$ va $\text{Var}[\tilde{h}_j | h_j] = \frac{p}{1-p}h_j^2$ ni isbotlang; $h_j = 2$ uchun dispersiya qancha?
- Adam + L2 va AdamW.** $\theta_0 = 1,5$, doimiy gradient $g = 0,2$, $\eta = 0,01$, $\lambda = 0,1$, $\beta_1 = 0,9$, $\beta_2 = 0,999$, $\epsilon = 10^{-8}$. Birinchi qadamdan keyingi θ_1 ni (a) Adam, (b) Adam(weight_decay=0.1) (L2), (c) AdamW(weight_decay=0.1) uchun toping.
- Batch normalization.** Bitta kanal, paket $z = (0, 2, 4, 10)$, $\gamma = 2$, $\beta = 1$, $\epsilon = 10^{-5}$ (hisobda $\epsilon \rightarrow 0$). O'qitish rejimidagi chiqishni toping. PyTorch qoidasi: `running ← (1 - m)running + m · stat`, $m = 0,1$, boshlang'ich `running_mean = 0`, `running_var = 1`, dispersiya uchun *siljimagan* ($n - 1$) baho olinadi. Bir qadamdan so'ng `running` statistikalar va `eval()` rejimida BN(7) ni hisoblang.
- Initialization.** (a) Linear(784,256): He (normal) va Xavier (normal) std. (b) Conv2d(64,128,3): fan_{in} , fan_{out} , He std va Xavier uniform chegarasi $a = \sqrt{6/(\text{fan}_{\text{in}} + \text{fan}_{\text{out}})}$. (c) Leaky ReLU (qiyalik a) uchun $v = \frac{2}{(1+a^2)n_{\text{in}}}$ ni keltirib chiqaring; $a = 0,2$, $n_{\text{in}} = 512$ da std? (d) PyTorch da `nn.Linear` ning standart initsializatsiyasi $U(-1/\sqrt{n_{\text{in}}}, 1/\sqrt{n_{\text{in}}})$: dispersiyasi He dan necha marta kichik?
- Warmup + cosine.** $\eta_{\text{max}} = 3 \cdot 10^{-3}$, $\eta_{\text{min}} = 10^{-5}$, $T_w = 5$, $T = 50$: $t < T_w$ da $\eta_t = \eta_{\text{max}}t/T_w$, $t \geq T_w$ da $\eta_t = \eta_{\text{min}} + \frac{1}{2}(\eta_{\text{max}} - \eta_{\text{min}})(1 + \cos \frac{\pi(t-T_w)}{T-T_w})$. $\eta_0, \eta_2, \eta_5, \eta_{20}, \eta_{50}$ ni toping.
- Adam va momentum.** (a) $\theta_0 = 1$, $\eta = 0,01$, $g_1 = 0,2$, $g_2 = -0,1$: $m_t, v_t, \hat{m}_t, \hat{v}_t$ va θ_2 ni hisoblang. Nega $g_2 < 0$ bo'lsa ham θ kamayishda davom etadi? (b) SGD momentum, $\mu = 0,9$, $g_t \equiv g$: u_5 va $\lim u_t$.

12. **Ansambl.** M ta model bashorati bir xil dispersiya σ^2 va juft korrelyatsiya ρ ga ega. $\text{Var}(\frac{1}{M} \sum f_k) = \rho\sigma^2 + \frac{1-\rho}{M}\sigma^2$ ni isbotlang. $\sigma^2 = 4$, $\rho = 0,25$: $M = 4$ va $M \rightarrow \infty$ uchun qiymat; dispersiya $\leq 1,2$ bo'lishi uchun eng kichik M ?

B. Konseptual savollar (13–16)

13. **Egri chiziqlar tashxisi.** Uch yurishning oxirgi o'quv/validatsiya yo'qotishlari: A (1,20; 1,25), B (0,02; 0,95), C (0,15; 0,19). Har biriga tashxis va birinchi davo. So'ng $v = (0,92; 0,71; 0,64; 0,66; 0,63; 0,65; 0,67; 0,68)$, $\text{patience} = 3$ uchun early stopping qaysi epoxadan so'ng to'xtaydi va qaysi epoxa (noldan) holati tiklanadi?
14. **Augmentation.** Gorizontaal aks ettirish va 180° burish qaysi masalada yorliqni saqlaydi: (a) MNIST raqamlari, (b) CIFAR-10, (c) ko'krak rentgeni (yurak chap tomonda), (d) sun'iy yo'ldosh tasvirlarida yer turi? Javobni asoslang.
15. **Xatolarni toping.** Quyidagi kodda kamida 5 ta metodologik xato bor. Har birini tuzating.

```
X = StandardScaler().fit_transform(X) # 1
Xtr, Xte, ytr, yte = train_test_split(X, y, random_state=42)
for ep in range(100):
    train_one_epoch(model, opt, Xtr, ytr) # ichida sched.step(); opt.step() # 2
    v = evaluate(model, Xte, yte) # 3: model.train() rejimida
    if v < best: best_state = v, model.state_dict() # 4
print("test:", evaluate(model, Xte, yte)) # 5
```

16. **Random search va grid.** Fazoning eng yaxshi $q = 5\%$ qismi maqbul. $n = 20$ random sinovda kamida bittasi maqbul bo'lish ehtimolligi? Ehtimollik $\geq 0,99$ uchun eng kichik n ? 3 ta giperparametr, har biri 4 qiymatli grid (64 sinov) har bir o'q bo'yicha nechta turli qiymat sinaydi, 64 ta random sinov-chi?

C. Kodli topshiriqlar (17–30)

Barcha kodli topshiriqlar quyidagi umumiy yordamchi kod bilan ishlaydi (uni `solutions.py` boshiga ko'chiring). Har bir topshiriq — `task<N>_...()` funksiyasi; chiqish satri `[N]` bilan boshlanadi.

```
import copy, math, random
import numpy as np, torch, torch.nn as nn, torch.nn.functional as F
from sklearn.datasets import load_digits, load_breast_cancer
from sklearn.model_selection import train_test_split
SEED = 42; torch.set_num_threads(1)
def set_seed(s=SEED): random.seed(s); np.random.seed(s); torch.manual_seed(s)
def digits_split(): # 60/20/20, stratified, X/16 in [0,1]
    X, y = load_digits(return_X_y=True); X = (X / 16.0).astype(np.float32)
    Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.2, stratify=y, random_state=SEED)
    Xtr, Xva, ytr, yva = train_test_split(Xtr, ytr, test_size=0.25, stratify=ytr, random_state=SEED)
    return Xtr, Xva, Xte, ytr, yva, yte
def add_label_noise(y, rate=0.3, seed=SEED): # rate*n indeksda yorliq <- randint(0,10)
    rng = np.random.default_rng(seed); y = y.copy()
    idx = rng.choice(len(y), int(rate * len(y)), replace=False)
    y[idx] = rng.integers(0, 10, len(idx)); return y
def mlp(sizes, p_drop=0.0, bn=False): # Linear -> [BatchNorm1d] -> ReLU -> [Dropout], oxiri Linear
    L = []
    for a, b in zip(sizes[:-2], sizes[1:-1]):
        L += [nn.Linear(a, b)] + ([nn.BatchNorm1d(b)] if bn else []) + [nn.ReLU()] \
            + ([nn.Dropout(p_drop)] if p_drop > 0 else [])
    return nn.Sequential(*L, nn.Linear(sizes[-2], sizes[-1]))
def train_epochs(model, opt, X, y, epochs, bs=32, augment=None, gen=None):
    for _ in range(epochs): # har epoxa: randperm(generator=gen), mini-paketlar, CE
        model.train(); perm = torch.randperm(len(X), generator=gen)
        for i in range(0, len(X), bs):
            idx = perm[i:i + bs]; xb = X[idx] if augment is None else augment(X[idx], gen)
            loss = F.cross_entropy(model(xb), y[idx])
            opt.zero_grad(); loss.backward(); opt.step()
```

Agar boshqacha aytilmagan bo'lsa: model qurishdan oldin `set_seed()`, `gen = torch.Generator().manual_seed(42)`; aniqlik `model.eval()` va `torch.no_grad()` ostida.

17. **Dropout Monte Carlo.** `task17_dropout(): set_seed(); x=torch.ones(1_000_000), nn.Dropout(p=0.3)` `train()` rejimida. Nollar ulushi, o'rtacha va dispersiyani (`unbiased=False`) chiqaring va 6-topshiriq formulasi bilan solishtiring; `eval()` rejimida $\max|y - x|$. Format: [17] zero ulushi = 0.xxxx mean = 0.xxxx var =

0.xxxx.

18. **Initialization va chuqurlik.** `task18_init_depth():` 20 ta `nn.Linear(256,256,bias=False)` + ReLU; kirish `torch.randn(1000,256)`. Uch sxema (har biri oldidan `set_seed()`): `kaiming_normal_(w, nonlinearity='relu')`, `xavier_normal_(w)`, `normal_(w, std=0.01)`. Birinchi qatlam `std(w)` va $l = 5, 10, 20$ qatlamdan keyingi faollik `std` ni jadval qilib chiqaring. Qaysi sxema signalni saqlaydi va nega?
19. **BatchNorm1d tekshiruvi.** `task19_batchnorm():` `nn.BatchNorm1d(1, momentum=0.1)`, `weight=2`, `bias=1`, kirish $z = (0, 2, 4, 10)^T$ (shakl 4×1). Train chiqishini qo'lda formula bilan, `running_mean`, `running_var` ni 8-topshiriq bilan solishtiring; `eval()` da `BN(7)` (4 xona).
20. **Adam, Adam + L2, AdamW.** `task20_adam_vs_adamw():` `nn.Parameter(torch.tensor([1.5]))`, yo'qotish $0,2\theta$ (doimiy gradient), `lr=0.01`; uch optimizator: `Adam(weight_decay=0)`, `Adam(weight_decay=0.1)`, `AdamW(weight_decay=0.1)`. 1- va 100-qadamdan keyingi θ (5 xona). Nega 100 qadamda Adam + L2 regularizatsiyasiz Adam dan kattaroq θ beradi?
21. **LR jadvali.** `task21_lr_schedule():` `SGD(nn.Linear(1,1).parameters(), lr=3e-3)` va `LambdaLR` bilan 10-topshiriqdagi warmup + cosine ni amalga oshiring; har qadamda avval `get_last_lr()` ni yozing, so'ng `opt.step()`, `sched.step()`. $t = 0, 2, 5, 20, 50$ uchun [21] $t=20$ `eta=2.252500e-03` formatida chiqaring.
22. **Early stopping (shovqinli yorliqlar).** `task22_early_stopping():` `digits_split()`; o'quv yorliqlariga `add_label_noise(ytr, seed=42)`, validatsiyaga `add_label_noise(yva, seed=43)`; `StandardScaler` faqat o'quv qismiga fit. `mlp([64,256,256,10])`, `Adam(lr=1e-3)`, `bs=32`, ko'pi bilan 200 epoxa, validatsiya cross-entropy si bo'yicha `patience=10`, eng yaxshi holat `copy.deepcopy(state_dict())`. Chiqish: to'xtagan va eng yaxshi epoxa (noldan), eng yaxshi va oxirgi val loss, oxirgi va tiklangan holatning test aniqligi (test yorliqlari toza).
23. **Regularizatsiya tadqiqoti.** `task23_regularization_study():` to'liq o'quv qismi ($n = 1077$), yorliqlar `add_label_noise(ytr)`; buzilgan yorliqlar ulushini chiqaring. `mlp([64,256,256,10])`, `AdamW(lr=1e-3)`, `bs=32`. Beshta usul: bazaviy (100 ep., `weight_decay=0`); dropout $p = 0,5$; `weight_decay=0.5`; Gauss shovqini augmentation (`xb + 0.2*torch.randn(xb.shape, generator=gen).clamp(0,1)`); bazaviy, atigi 10 epoxa. Har biri uchun o'quv (shovqinli yorliqlar bo'yicha) va test aniqligi. Qaysi usul yodlashni to'xtatadi?
24. **L1 va L2 yo'li.** `task24_l1_l2_path():` `load_breast_cancer`, `test_size=0.3`, `stratify=y`, `random_state=42`, `StandardScaler` (o'quvda fit). $C \in \{0,01; 0,1; 1; 10; 100\}$ uchun `LogisticRegression(C=C, max_iter=10000)` (L2) va `LogisticRegression(C=C, l1_ratio=1.0, solver='liblinear', max_iter=10000)` (L1). Diqqat: `scikit-learn 1.8` da `penalty` parametri eskirgan (deprecated), uni ishlatmang. Har C uchun $\|w\|_2$, nol koeffitsiyentlar soni, test aniqligi.
25. **Bias-variance simulyatsiyasi.** `task25_bias_variance():` `rng=np.random.default_rng(42)`, $f(x) = \sin 2\pi x$, $\sigma = 0,3$, $n = 30$, $R = 200$ takror, $x \sim U(0,1)$, baholash to'ri `np.linspace(0.1,0.9,50)`. Darajalar 1,3,9 (`np.polynomial.Polynomial.fit`). Har takrorda (tartib: x , o'quv shovqini, test shovqini) yangi test nishonlari bilan MSE, bias^2 , variance, $\sigma^2 + \text{bias}^2 + \text{variance}$ va empirik test MSE ni solishtiring.
26. **API tekshiruvlari.** `task26_api_checks():` (a) `F.cross_entropy(z, t, label_smoothing=0.1)` ni 5-topshiriq bilan va qo'lda formula bilan solishtiring; (b) grad $(2, -1, 2)$ li parametrga `nn.utils.clip_grad_norm(..., max_norm=1.5)`: qaytgan norma va yangi gradient.
27. **Ma'lumot sizishi.** `task27_leakage():` `rng=default_rng(42)`, X — 100×5000 standart normal, y — `rng.permutation(np.repeat([0,1],50))` (yorliq *tasodifiy*). `StratifiedKfold(5, shuffle=True, random_state=42)`. (a) `SelectKBest(f_classif,k=20)` ni butun X ga moslab, keyin `cross_val_score(LogisticRegression(max_iter=1000))`; (b) xuddi shu, lekin `make_pipeline` ichida. O'rtacha CV aniqliklari. Qaysi raqam haqiqatni aytadi?
28. **Qaysi qatlam ko'chiriladi (transfer learning).** `task28_transfer():` manba — 0–4 raqamlar ($X/16$), nishon — 5–9 raqamlar (yorliq $y - 5$), nishondan faqat 50 ta o'quv namunasi (`train_size=50`, `stratify`, `random_state=42`). Tayanch `Sequential(Linear(64,128),ReLU,Linear(128,64),ReLU)` + bosh `Linear(64,5)` manbada 30 epoxa (`Adam 1e-3`). Muzlatilgan belgilarda `LogisticRegression(C=1, max_iter=5000)` bilan linear probing: xom piksellar; 1-qatlam (128); 2-qatlam (64); tasodifiy tayanchning 1-qatlami (`set_seed(43)`). Test aniqligi va o'qitiladigan parametrlar soni. Natijani izohlang.
29. **Ansambl.** `task29_ensemble():` `digits_split()` o'quv qismidan 300 ta (`stratify`, `random_state=42`); `mlp([64,64,10])`, `Adam(1e-3)`, 40 epoxa, urug'lar 42,...,46 (har biri uchun `set_seed` va `gen`). Har bir model, o'rtacha va softmax o'rtachasi bo'yicha ansamblning test aniqligi. 12-topshiriq bilan bog'lang.
30. **BatchNorm va chuqurlik.** `task30_bn_depth():` `mlp([64]+[64]*10+[10], bn=False/True)`, standart PyTorch initsializatsiyasi, `SGD(lr=0.05, momentum=0.9)`, 15 epoxa, to'liq o'quv qismi. 1-, 5-, 15-epoxadan keyingi o'quv yo'qotishi (`eval()` da, butun o'quv qismi) va test aniqligi. 9(d)-topshiriq natijasi bilan izohlang.

Hujjatlar (documentation)

- <https://docs.pytorch.org/docs/stable/generated/torch.nn.Dropout.html>
- <https://docs.pytorch.org/docs/stable/nn.init.html>

- <https://docs.pytorch.org/docs/stable/generated/torch.nn.BatchNorm1d.html>
- <https://docs.pytorch.org/docs/stable/generated/torch.optim.Adam.html>, <https://docs.pytorch.org/docs/stable/generated/torch.optim.AdamW.html>
- https://docs.pytorch.org/docs/stable/generated/torch.optim.lr_scheduler.LambdaLR.html
- https://docs.pytorch.org/docs/stable/generated/torch.nn.functional.cross_entropy.html
- https://docs.pytorch.org/docs/stable/generated/torch.nn.utils.clip_grad_norm_.html
- https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_digits.html
- https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html
- https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html
- https://scikit-learn.org/stable/modules/generated/sklearn.feature_selection.SelectKBest.html
- https://scikit-learn.org/stable/modules/generated/sklearn.pipeline.make_pipeline.html
- <https://numpy.org/doc/stable/reference/generated/numpy.polynomial.polynomial.Polynomial.fit.html>
- <https://docs.python.org/3/library/copy.html>